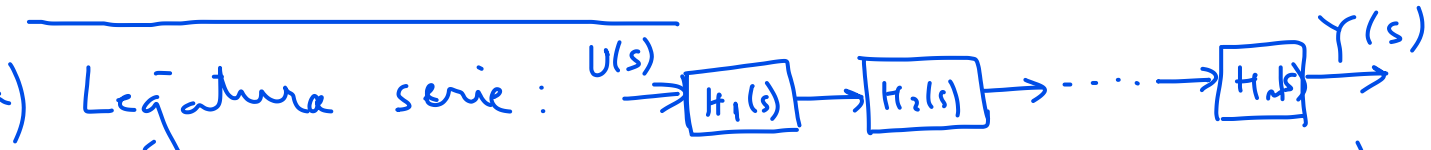
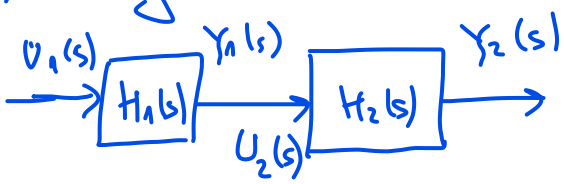
Conexiunea sistemelor

a) Legătura serie:





$U(s) = U_1(s)$

$Y_1(s) = U_2(s)$

$Y_2(s) = Y(s)$

$H_1(s) = \frac{Y_1(s)}{U_1(s)}$

$H_2(s) = \frac{Y_2(s)}{U_2(s)}$

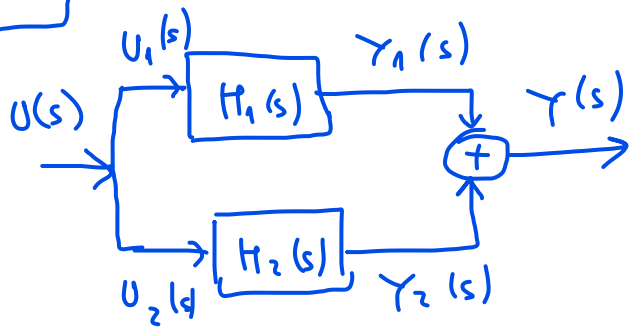


$$H_{serie}(s) = \frac{Y(s)}{U(s)} = \frac{Y_2(s)}{U_1(s)} = \frac{H_2(s) \cdot \cancel{U_2(s)}}{\cancel{Y_1(s)}} = H_1(s) \cdot H_2(s)$$

$$H_1(s)$$

$$\boxed{H_{serie}(s) = \prod_{i=1}^{n} H_i(s)}$$

Legatura paralel.



$U(s) = U_1(s) = U_2(s)$

$Y(s) = Y_1(s) + Y_2(s)$

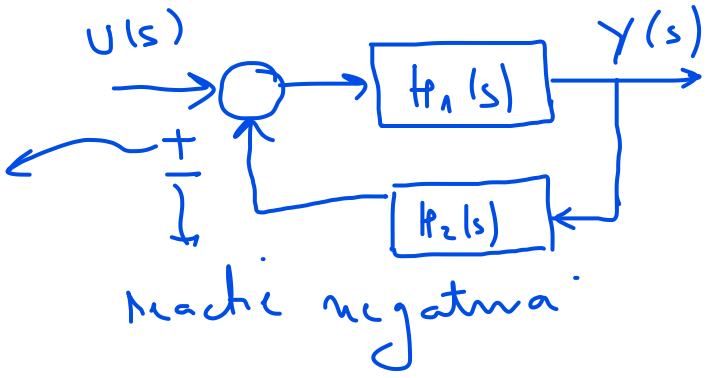
$H_1(s) = \frac{Y_1(s)}{U_1(s)}$ $\qquad H_2(s) = \frac{Y_2(s)}{U_2(s)}$

$$H_{paralel}(s) = \frac{Y(s)}{U(s)} = \frac{Y_1(s) + Y_2(s)}{U(s)} = \frac{H_1(s) \cdot \cancel{U_1(s)} + H_2(s) \cdot \cancel{U_2(s)}}{\cancel{U(s)}}$$

$$= H_1(s) + H_2(s)$$

$$\boxed{H_{paralel}(s) = \sum_{i=1}^{n} H_i(s)}$$

Legatura reactie



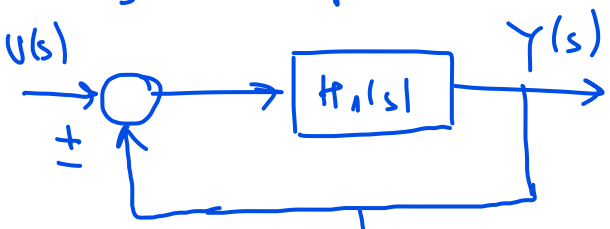
reactie pozitiva

reactie negativa

$$H_{reactie}(s) = \frac{H_1(s) \cdot H_2(s)}{1 \mp H_1(s) \cdot H_2(s)}$$

semn opus tipului de reactie

Obs



consideram functie de transfer = 1

$$H_{reactie}(s) = \frac{H_1(s)}{1 \mp H_1(s)}$$

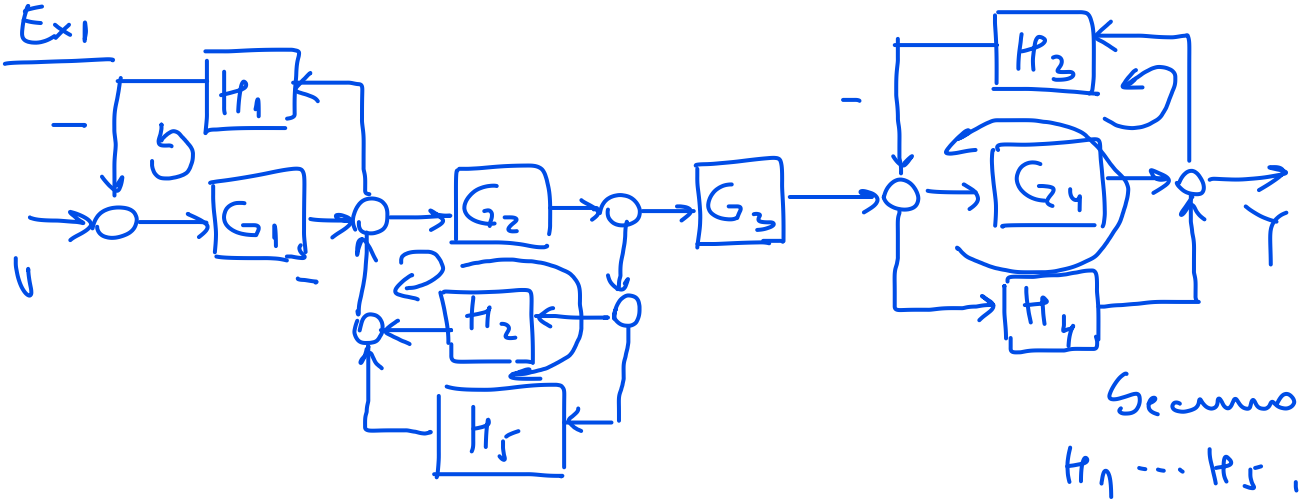
Obs. ① Legatura serie si paralel este stabila doar daca toate elementele (blocurile) din conexiune au aceasta proprietate

② Legatura reactie nu are aceasta proprietate

=> este folosita pt a stabiliza bloc de pe calea directa.

Regula de manipulare a functiilor de transfer a lui MASON.

Ex 1



Se cere: $H_1 \dots H_5$; $G_1 \dots G_4$

Se cere $\frac{Y}{U}$

a) Căi directe de la U la Y.

$G_{uy_1} = G_1 G_2 G_3 G_4$

$G_{uy_2} = G_1 G_2 G_3 H_4$

b) Bucle. ($j = 1$)

$P_1(1) = -G_1 H_1$ $\quad P_3(1) = -G_2 H_5$

$P_2(1) = -G_2 H_2$ $\quad P_4(1) = -G_4 H_3$

$P_5(1) = -H_4 H_3$

c) Perechi de bucle care nu se ating ($j = 2$)

$P_1(2) = G_1 G_2 H_1 H_2$ $\quad P_5(2) = G_2 G_4 H_2 H_3$

$P_2(2) = G_1 G_2 H_1 H_5$ $\quad P_6(2) = G_2 H_2 H_3 H_4$

$P_3(2) = G_1 G_4 H_1 H_3$ $\quad P_7(2) = G_2 G_4 H_3 H_5$

$P_4(2) = G_1 H_1 H_3 H_4$ $\quad P_8(2) = G_2 H_3 H_4 H_5$

d) Triplete de bucle care nu se ating ($j = 3$)

$P_1(3) = -G_1 G_2 G_4 H_1 H_2 H_3$ $\quad P_3(3) = -G_1 G_2 G_4 H_1 H_3 H_5$

$P_2(3) = -G_1 G_2 H_1 H_2 H_3 H_4$ $\quad P_4(3) = -G_1 G_2 H_1 H_3 H_4 H_5$

e) Cvartele de bucle care nu se ating ($j=4$)

$P_1(4) = 0$.

Calculam $\underline{\Delta}$ - parametrul ce caracterizează global schema:

$$\Delta = 1 + (-1)^1 \cdot [P_1(1) + P_2(1) + P_3(1) + P_4(1) + P_5(1)]$$

$$+ (-1)^2 \cdot [P_1(2) + P_2(2) + \dots + P_8(2)]$$

$$+ (-1)^3 \cdot [P_1(3) + \dots + P_4(3)] + (-1)^4 \cdot 0$$

$$= 1 + [G_1 H_1 + \dots] + [G_1 G_2 H_1 H_2 + \dots]$$

$$+ [G_1 G_2 G_4 H_1 H_2 H_3]$$

Calculam câte un parametru Δ_i pentru fiecare cale directa Guy_i:

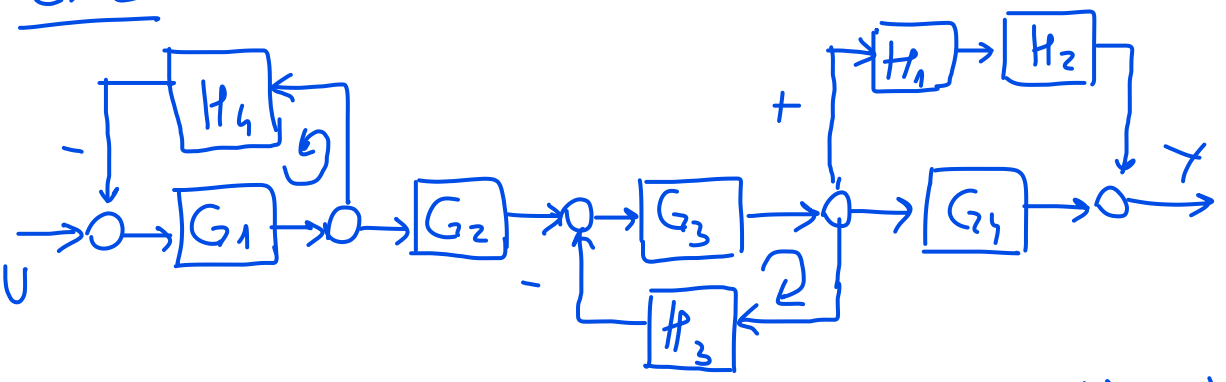
$$\Delta_1 = 1 + (-1)^1 \cdot [P_3(1)] + (-1)^2 \cdot 0 = 1 + H_3 H_4$$

+ coef căii Guy_1 ptcă $\underline{\text{nu}}$ atinge calea Guy_1

$$\Delta_2 = 1 + (-1)^1 \cdot [P_4(1)] + (-1)^2 \cdot 0 = 1 + G_4 H_3$$

$$\frac{y}{x} = \frac{Guy_1 \cdot \Delta_1 + Guy_2 \cdot \Delta_2}{\Delta}$$

Ex 2



Fcare $\frac{Y}{U}$

a) Căi directe de la U la Y.

$G_{uy_1} = G_1 G_2 G_3 G_4$

$G_{uy_2} = G_1 G_2 G_3 H_1 H_2$

b) Bucle $(j=1)$

$P_1(1) = -G_1 H_4$

$P_2(1) = -G_3 H_3$

c) Perechi de bucle care nu se ating $(j=2)$

$P_1(2) = G_1 G_3 H_3 H_4$

d) Triplete de bucle care nu se ating $(j=3)$

$P_1(3) = 0$

$\Delta = 1 + (-1)^1 \cdot [P_1(1) + P_2(1)] + (-1)^2 \cdot P_1(2) + (-1)^3 \cdot 0$

$= 1 + G_1 H_4 + G_3 H_3 + G_1 G_3 H_3 H_4$

$\Delta_1 = 1 + (-1)^1 \cdot [0] = 1$

$\Delta_2 = 1 + (-1)^1 \cdot [0] = 1$

$\Rightarrow \frac{Y}{X} = \frac{G_{uy_1} \cdot \Delta_1 + G_{uy_2} \cdot \Delta_2}{\Delta} =$

$= \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_3 H_1 H_2}{1 + G_1 H_4 + G_3 H_3 + G_1 G_3 H_3 H_4}$